

Problem

When the periodic voltage $2 + 6 \sin \omega t$ is applied to a $1\text{-}\Omega$ resistor, the integer closest to the power (in watts) dissipated in the resistor is:

- (a) 5
- (b) 8
- (c) 20
- (d) 22
- (e) 40

Step-by-step solution

Step 1 of 1

$$\begin{aligned} p &= a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 + b_n^2 \\ &= 2^2 + \frac{1}{2} \times 6^2 \\ &= 4 + 18 \\ &= 22 \text{ Ans.} \end{aligned}$$